Suggested problems: 1-11 odd, 41, 43

Problem 1: If $f(x, y) = x^4y^2 - x^3y$, find:

(a) $f_x(x, y)$

(b) $f_y(x, y)$

(c) $f_{xx}(x, y)$

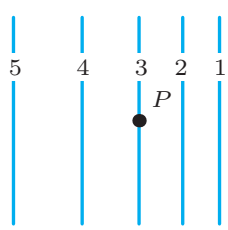
(d) $f_{xy}(x, y)$

(e) $f_{yx}(x, y)$

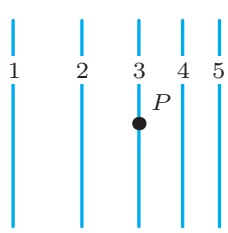
(f) $f_{yy}(x, y)$

Problem 2: For each of the contour plots below, decide if the signs of the various partial derivatives at the point P are positive or negative, or if the derivatives are zero. The derivatives are $f_x(P)$, $f_y(P)$, $f_{xx}(P)$, $f_{yy}(P)$, $f_{xy}(P)$.

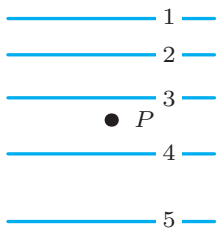
(a)



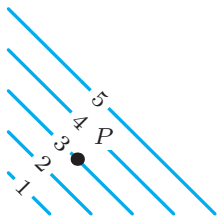
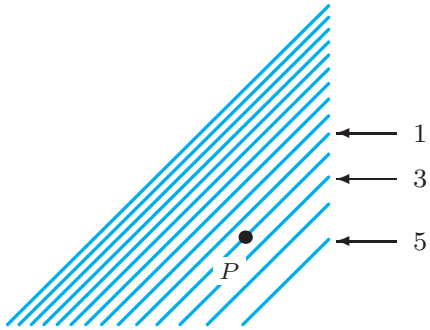
(b)



(c)



(d)



(e)